EXCEL: PAGE LAYOUT, PRINTING & PDF PUBLISHING

Complete practical handbook | Lessons 37-41 | Excel desktop (Windows)

A consolidated reference for every topic covered in the printing section: exact ribbon paths, print settings, examples, expected outcomes, common problems, and reference tables. Practice assignments are deliberately excluded.

Contents

Lesson	Topics
37	Orientation, A4 paper, margins, print preview, scaling, PDF export
38	Set/Clear/Add Print Area, Print Selection, exclude slicers
39	Manual/automatic page breaks, preview, reset breaks, multiple print areas
40	Wide and long datasets, repeated headers and columns, gridlines
41	Page Setup: Page, Margins, Header/Footer, Sheet, errors, final publishing

Core navigation at a glance

What to do	Path
Print preview	Ctrl + P or File > Print
Page settings	Page Layout > Page Setup (dialog launcher)
Print range	Page Layout > Print Area
Page breaks	Page Layout > Breaks
Repeated headers	Page Layout > Print Titles
Export	File > Export > Create PDF/XPS

Worked dataset used throughout

Workbook: Excel_Complete_Practice.xlsx. The example data are illustrative, consistent with the lessons. Enter each block into the named sheet as shown; results describe printing layout, not mathematical analysis.

Sheet / range	Example content
37_PageLayout, A1:G7	Agent, Product, Order Type, Sales, Orders, Avg CSAT, Discount; six agent/product rows.
38_PrintArea, A1:B5	Sales summary: total sales 69,500; orders 47; avg CSAT 7.34; avg discount 9.28%.
38_PrintArea, D1:E4	Agent summary: Adrian 20,500; Albion 23,500; Roche 25,500.

Sheet / range	Example content
38_PrintArea, G1:H4	Product summary: Pizza 41,000; Burger 28,500; grand total 69,500.
39_PageBreaks, A1:C5 and A9:C12	Sales metrics / targets in first block; agent-wise sales and orders in second block.

LESSON 37 | PAGE LAYOUT, PREVIEW, SCALING & PDF

37.1 What Page Layout controls

The Page Layout ribbon determines how worksheet content will fit on paper or in an exported PDF. It does not change the underlying cell values.

Setting	Function	When useful
Orientation	Portrait (tall) or Landscape (wide)	Landscape for many dashboard columns
Size	Select A4, Letter, other supported paper sizes	A4 when the output requires 210 x 297 mm
Margins	Normal, Wide, Narrow, Custom	Narrow to provide more printable room
Scaling	Control printed size and page count	Small summaries vs long datasets
Print Preview	Inspect pagination before printing	Find missing columns, tiny text, blank pages

37.2 Example dataset

Agent	Product	Type	Sales	Orders	CSAT	Discount
Adrian	Pizza	Online	12,000	8	8.2	5%
Adrian	Burger	Visit	8,500	6	6.2	12%
Albion	Burger	Online	9,500	7	5.8	15%
Albion	Pizza	Visit	14,000	10	8.5	6%
Roche	Pizza	Online	15,000	9	7.8	8%
Roche	Burger	Visit	10,500	7	6.8	11%

37.3 Print a small summary on one A4 landscape page

1. Open 37_PageLayout and press Ctrl + P (File > Print).

2. Change Portrait Orientation to Landscape Orientation.

3. Change paper size to A4 (the lesson starts with Letter-sized paper).

4. Set Margins to Narrow if columns do not fit.

5. Open Scaling and select Fit Sheet on One Page.

6. Inspect the preview to ensure the report is readable and no column is missing.

Expected result: This small report fits onto one landscape A4 page, subject to the worksheet’s widths and printer/PDF settings.

37.4 Understand scaling options

Scaling option	What happens	Preferred situation
No Scaling	Keeps printing scale rather than fitting automatically	When original sizing is appropriate
Fit Sheet on One Page	Fits all rows and columns to one page; can shrink excessively	Small summary/dashboard
Fit All Columns on One Page	One page wide, as many pages tall as needed	Long tables with many columns
Fit All Rows on One Page	One page tall; columns can continue to other pages	Short, unusually wide data

37.5 PDF publishing methods

1. Export: File > Export > Create PDF/XPS Document > Create PDF/XPS > choose filename > Publish.

2. Print to PDF: Ctrl + P > select Microsoft Print to PDF as printer > Print > enter filename and Save.

3. Verify the produced PDF page count, legibility, and included worksheets/ranges.

Important: Landscape can increase rather than decrease page count for tall tables. A4 size and margins also alter pagination. Always verify the preview rather than assuming the number of pages will fall.

LESSON 38 | PRINT AREA, PRINT SELECTION & SLICERS

38.1 Print Area and Print Selection are different

Feature	Print Area	Print Selection
Purpose	Saved region for repeated printing	One-time current cell selection
Where	Page Layout > Print Area	File > Print > Print Selection
Persistence	Saved with workbook	Depends on current selection
Example	Always print A1:E5	This time print D1:E4

38.2 Example: three summaries on one sheet

Location	Contents	Sample results
A1:B5	Overall summary	Sales 69,500; Orders 47; CSAT 7.34; Discount 9.28%
D1:E4	Agents	Adrian 20,500; Albion 23,500; Roche 25,500
G1:H4	Products	Pizza 41,000; Burger 28,500; Total 69,500

38.3 Set a persistent Print Area

1. Select A1:E5 on sheet 38_PrintArea (sales + agents; excludes product summary).

2. Choose Page Layout > Print Area > Set Print Area.

3. Press Ctrl + P, and inspect the preview.

4. For layout, select Landscape, A4, a suitable margin and scaling setting.

Expected result: Only the A1:E5 area prints, excluding G1:H4. The named print area is stored with the workbook.

38.4 Clear and add print areas

1. To remove the saved limitation, select Page Layout > Print Area > Clear Print Area. Source data remains intact.

2. To create separated areas, select A1:B5 > Set Print Area; then select G1:H4 > Add to Print Area.

3. Preview the output; separate nonadjacent areas typically print on separate pages.

38.5 Print only the current selection

1. Select D1:E4 (Agent Summary).

2. Press Ctrl + P.

3. Under Settings, replace Print Active Sheets with Print Selection.

4. Confirm that only the selected agent table appears; print or export as needed.

Important: If you select one cell before Print Selection, only that cell or a tiny area may be included. The selection overrides the usual saved-print-area workflow for this print operation.

38.6 Slicers and objects in a printed report

Slicers outside the defined print rectangle are generally excluded. A slicer overlapping the chosen rectangle can still print. Move it outside the print area or, where supported, open its Size & Properties and disable Print object. Some controls vary by Excel version/object type.

LESSON 39 | PAGE BREAKS & MULTIPLE PRINT AREAS

39.1 Automatic versus manual breaks

Type	Who controls it	What to expect
Automatic	Excel from paper, margins, content and scale	Displayed as dashed blue boundaries in Page Break Preview
Manual	You choose a row/column split	Usually displayed as solid blue boundaries

39.2 Dataset and desired output

39_PageBreaks: Sales Summary occupies A1:C5: Sales 69,500 vs target 70,000; Orders 47 vs 50; CSAT 7.34 vs 7.5; Discount 9.28% vs 8%. Agent Summary occupies A9:C12: Adrian 20,500 / 14; Albion 23,500 / 17; Roche 25,500 / 16. Desired output: first summary Page 1, second summary Page 2.

39.3 Insert a horizontal manual page break

1. Select A9, which is the start of the next section.

2. Choose Page Layout > Breaks > Insert Page Break.

3. Open View > Page Break Preview to inspect the boundary above row 9.

4. Press Ctrl + P; verify the summaries fall on separate pages.

Important: To insert a vertical break before column E, select column E and use Insert Page Break. Fit-to-one-page scaling can override manual page breaks.

39.4 Remove or reset breaks

1. To remove one horizontal break, select the row directly below it (for example A9), then Page Layout > Breaks > Remove Page Break.

2. To clear all manually inserted breaks from the worksheet, choose Page Layout > Breaks > Reset All Page Breaks.

3. Return to View > Normal when finished.

39.5 Multiple print areas versus page breaks

Situation	Better method
Print two separated summaries and omit intervening worksheet cells	Set first Print Area; Add to Print Area for the second
Split one continuous report between rows 8 and 9	Manual page break above row 9
Print only the selected agent block once	Print Selection
Unexpected extra page or strange split	Page Break Preview; inspect selection, breaks, scaling, merged cells

39.6 Troubleshooting

If a break seems ignored, check Page Setup > Page > Scaling: an aggressive Fit to 1 page setting can suppress manual pagination. If a second blank page appears, review Page Break Preview, the selected print area, merged headings and any accidental blank/remote cells within the printable range.

LESSON 40 | LARGE DATASETS, PRINT TITLES & GRIDLINES

40.1 Problem: wide and long datasets

A dataset with approximately 700 rows and 10-12 columns may print as many pages, with half the columns on later pages and headers missing after Page 1. Switching to Landscape alone does not guarantee a better result. Fitting the entire sheet on one page makes text too small.

40.2 Fit columns to one page width, not all rows

1. Open the full dataset and press Ctrl + P.

2. Set Landscape and A4 where suitable; use Narrow margins if required.

3. Choose Fit All Columns on One Page.

4. Inspect page 1 and later pages: the columns remain together, while rows extend onto further pages.

Expected result: A readable long table whose columns fit the page width; multiple pages continue down the rows. If 12 very wide columns remain tiny, reduce widths or choose larger paper rather than forcing them excessively.

40.3 Repeat header rows with Print Titles

1. Go to Page Layout > Print Titles.

2. On the Sheet tab, find Rows to repeat at top.

3. Enter $1:$1 to repeat the first row (or $1:$2 for the first two rows).

4. Click OK; open Print Preview and inspect Page 2 to confirm headers repeat.

Important: The repeated-header selection uses entire rows, not individual header cells. Print Titles is for printing; it is different from Freeze Panes, which controls scrolling in the worksheet.

40.4 Repeat columns and print gridlines

Need	Setting	Example
Repeat leftmost identifying column	Page Layout > Print Titles > Sheet > Columns to repeat at left	$A:$A
Print worksheet gridlines	Page Layout > Sheet Options > Gridlines > Print; or Page Setup > Sheet > Gridlines	Gridlines appear in printed output
Print row/column letters and numbers	Page Setup > Sheet > Row and column headings	A, B, C / 1, 2, 3 in print

40.5 Gridlines versus borders

Gridlines are Excel’s built-in cell guides and need an explicit print setting. Borders are cell formatting and print as styled. Gridline visibility in the preview may differ from the worksheet’s normal display; inspect the PDF for legibility.

LESSON 41 | ADVANCED PAGE SETUP & FINAL PUBLISHING

41.1 Open the Page Setup dialog

Go to Page Layout and click the small dialog launcher in the Page Setup group. The dialog contains Page, Margins, Header/Footer, Sheet tabs.

Tab	Key settings
Page	Portrait/Landscape, paper size, Adjust to % or Fit to page width/height, first page number
Margins	Top/bottom/left/right margins, header/footer spacing, Center on page horizontally/vertically
Header/Footer	Custom header or footer; page number, total pages, date, filename, worksheet name
Sheet	Print area, repeated titles, gridlines, black-and-white, draft quality, headings, errors, page order

41.2 Center a printed summary

1. Open Page Layout > Page Setup launcher > Margins.

2. Under Center on page, check Horizontally to center left to right.

3. Optionally check Vertically to center top to bottom.

4. Click OK and inspect Print Preview.

41.3 Headers, footers and error display

1. For page numbers or a report title, use Page Setup > Header/Footer > Custom Header/Custom Footer. Add items to left, center or right sections.

2. For Excel errors, open Page Setup > Sheet > Cell errors as. Choose Displayed, Blank, --, or #N/A.

3. Choose Displayed if stakeholders must see the actual worksheet errors; choose #N/A when a consistent print marker is preferred. This does not fix source formulas.

41.4 Prepare an interactive dashboard for PDF

1. Select only the summary cells; use Page Layout > Print Area > Set Print Area.

2. Move any slicer overlapping the print area; leave slicers outside the selected print rectangle or disable printing for the object when supported.

3. Choose Landscape, A4, and an appropriate margin.

4. For a compact dashboard, set Fit to 1 page wide by 1 tall; do not use this if text becomes unreadable.

5. Optionally use Center on page > Horizontally. Leave gridlines off for a polished dashboard unless required.

6. Open Ctrl + P, review all previews, and publish via File > Export > Create PDF/XPS or Microsoft Print to PDF.

7. Open the exported PDF and confirm that there are no missing columns, overlapping slicers, or unintentional pages.

41.5 Final settings reference

Document type	Typical layout	Main cautions
Single dashboard	Landscape, A4, suitable margins, Fit to 1 x 1, optional horizontal centering	Check readability; exclude print objects
Long transaction table	Landscape if wide; Fit All Columns on One Page; repeat headers	Do not compress hundreds of rows onto one page
Two isolated summaries	Separate print areas	They generally produce separate printed pages
One continuous report split into sections	Manual page break	Check scaling before assuming breaks will be honoured

41.6 Troubleshooting checklist

Symptom	Check / fix
Columns are missing	Print Area bounds, orientation, scaling, actual column widths
Pages are blank or split oddly	Selected cells, page breaks, stray formatted cells, merged headings
Headers missing on page 2	Print Titles > Rows to repeat at top
Slicer appears in PDF	Object overlaps Print Area; reposition it or disable print object
Everything looks tiny	Avoid Fit Sheet on One Page for large datasets
Error cells print differently	Page Setup > Sheet > Cell errors as
Printed data hugs top-left	Page Setup > Margins > Center on page

FAST REFERENCE | ALL ESSENTIAL PATHS

Task	Ribbon path or shortcut
Preview	Ctrl + P
Portrait / Landscape	Page Layout > Orientation
A4	Page Layout > Size
Margins	Page Layout > Margins
Fit columns	File > Print > Scaling > Fit All Columns on One Page
Set, clear or add print area	Page Layout > Print Area
Selected cells only	Select range > Ctrl + P > Print Selection
Insert/remove/reset breaks	Page Layout > Breaks
View page boundaries	View > Page Break Preview
Repeat first row	Page Layout > Print Titles > Sheet > $1:$1
Repeat column A	Page Layout > Print Titles > Sheet > $A:$A
Print gridlines	Page Layout > Sheet Options > Gridlines > Print
Center content	Page Setup > Margins > Center on page
Error print appearance	Page Setup > Sheet > Cell errors as
PDF export	File > Export > Create PDF/XPS

Scope: Consolidated from Lessons 37-41 of the Excel course. Includes covered explanations, sample data, steps, expected outcomes, practical cautions, and troubleshooting. Practice assignments omitted.